

Does the $d_{\text{mem}}=128$ dual-stream pair reproduce the Luo–Maunsell sensitivity/criterion dissociation?

A frozen-policy SDT assay of two partially trained, cost-rescued policies

Experiment report

18 August 2026

Abstract

Two RunPod training runs of the $d_{\text{mem}}=128$ dual-stream counterphased sensitivity design were terminated early for cost and their checkpoints rescued at iterations 18,799 (high-priority location 0) and 15,849 (location 3) of a 20,000-iteration contract. Re-running the repository’s frozen-policy signal-detection assay on both policies gives a clear split verdict. The *sensitivity* component of the Luo–Maunsell (LM) effect is **reproduced**: the counterphased difference-in-differences in d' is positive with a bootstrap CI excluding zero at all three evaluated difficulties ($\Delta d'_{\text{DiD}} = +0.23$ to $+0.32$). The *specificity* component — criterion invariance — is **not reproduced**: the criterion cross-effect is $\Delta c_{\text{DiD}} \approx +0.23$, whose CI extends past the preregistered ± 0.2 equivalence bound, so the strict dissociation test fails. The central new finding is *why* it fails. The observed criterion shift matches, to within 0.03, the reward-optimal criterion implied by the task’s own hit:correct-rejection ratios (0.7 at the high-priority location versus 1.1 at the low). The agents are not miscalibrated; they are sitting on the optimum that the reward schedule specifies. Because that optimum is $\Delta c = 0.452/d'$, the ± 0.2 target is unreachable unless $d' \gtrsim 2.26$ at both locations, whereas these policies achieve $d' = 1.67\text{--}2.10$. Doubling recurrent width did improve criterion *calibration* — the $d_{\text{mem}}=64$ parent overshoot the reward optimum by $2.1\times$, while this pair brackets it — but no amount of memory or training can satisfy a specificity target that the uncorrected reward table forbids. The design must be criterion-titrated before it can test the LM dissociation.

1 Scope, and what “the LM results” means here

“LM” is read throughout as Luo & Maunsell (2015, *Neuron* 86:1182–1188), the source protocol named in the task module `envs/luo2015.py`. The effect being chased is their dissociation between two components of attention: a spatial manipulation that changes detection *sensitivity* (d') should leave the decision *criterion* (c) essentially unchanged, and vice versa.

The repository encodes this as two preregistered targets (`latest_full_battery_manifest.json`):

- **sensitivity lineage** — positive counterphased d' difference-in-differences, with the criterion cross-effect inside an equivalence bound of $|\Delta c| \leq 0.2$;
- **criterion lineage** — negative counterphased c difference-in-differences, with the d' cross-effect inside the same bound.

Scope limit. Both rescued runs are *sensitivity* cells. The experiment package states plainly that “criterion cells are unsupported” in this design. This report can therefore address only the first half of the dissociation. Establishing a double dissociation would additionally require the criterion lineage, which was never run in this family. Nothing below should be read as evidence about the criterion-manipulation half.

A second scope limit is inherited from the experiment manifest itself, whose `claim_scope` field reads

```
memory_capacity_followup_in_uniform_change_noisy_
memory_extension_not_direct_luo_maunsell_replication
```

and which warns that the “fixed H:CR ratios directly pressure criterion.” That warning turns out to be the whole story (Section 4.4).

2 Provenance of the two runs

Both pods ran the identical launch contract from

```
experiments/luo2015_episodic/fresh_dualstream_dmem128_grid2_
memnoise0075_gamma100_bc000_curriculum_sensitivity_runpod
```

differing only in the `--high-loc` flag. Fresh seed-0 initialisation, 2×2 token grid, xLSTM cell with `crossattn1` feedback, separate actor and critic streams with independent recurrent state and independent JEPA branches at coefficient 0.5 each, teacher EMA 0.996, $\gamma = 1.0$, mnemonic-noise SD 0.075, sensory jitter 5° , BC coefficient 0, reward scale 1/3, shrinking- θ curriculum from 65° (window 1000 valid SDT trials, threshold 0.85, step 3° , floor 8°).

Table 1: Status of the two runs at rescue. The contract was 20,000 iterations; neither reached it. “Last observed” is the final successful probe by the local monitoring watchdog, after the rescue snapshot was taken.

	high-priority loc 0	high-priority loc 3
Pod ID	5ic3h9bmujytt2	kmmqwwmd4uaweu
Rescued checkpoint iteration	18,799	15,849
Metrics rows captured	18,840	15,888
Last iteration observed	19,516	15,986
Fraction of contract	94%	79%
Terminal curriculum θ	47°	38°
Final rolling correct rate	0.805	0.823
Best rolling correct rate	0.898	0.890
Checkpoint SHA-256 verified	yes	yes

Two honest caveats about the artifacts:

1. **The checkpoints are partial.** The repository’s terminal validators (`validate_checkpoint`, `validate_curriculum_checkpoint`) assert `iter == 19999` and were therefore bypassed. Every other element of the measurement contract was kept. Results below are frozen-policy measurements of non-terminal policies, not terminal replication evidence.
2. **A later checkpoint was lost.** The snapshot was taken at 04:44 UTC; the runs continued during download and verification, and the watchdog last saw a checkpoint at iteration 19,499 for the loc-0 cell before the pods were terminated at approximately 04:56 UTC. Roughly 700 further iterations of the loc-0 lineage exist only as now-deleted pod state. The analysis uses iteration 18,799.

The evaluation imports the model and environment definitions from the *rescued pod source tree*, not the working copy: `model.py`, `paper_encoder.py`, `ppo.py` and `train_rl.py` in the local repository have been modified since launch, whereas `envs/luo2015.py`, `envs/base.py` and `luo2015_analysis/luo2015_core.py` are byte-identical to the deployed versions.

3 Task and measurement contract

Two Gabors sit at diametrically opposed cells (0 = top-left, 3 = bottom-right) of a 2×2 grid, each with an independent orientation drawn from $\text{Uniform}[0^\circ, 180^\circ)$. There is no value cue. After a delay one location is tested, either unchanged or offset by a signed change drawn from $\text{Uniform}(-\theta, +\theta)$. A declaration during the first-test window (frames 3–4) is a hit or a false alarm; withholding is verified by a required declaration to a guaranteed-changed second test at frame 6, and failures of that check are excluded from the SDT denominators, as in the paper. $d' = z(\text{HR}) - z(\text{FA})$ and $c = -\frac{1}{2} [z(\text{HR}) + z(\text{FA})]$, with rates clamped to $[1/2N, 1 - 1/2N]$.

Measurement followed the repository assay exactly: one deterministic balanced trial bank shared by both policies, 2,000 trials per change-status per location (8,000 trials per bank), trained mnemonic noise enabled, sampled categorical policy actions, and 5,000-draw bootstrap confidence intervals. Evaluation ran on the exact rescued source under PyTorch 1.13.1/CUDA on an RTX 3070; all six rescued checkpoints were verified to deserialise with finite weights before use.

Because the two lineages stalled at different curriculum depths, three evaluation branches were run: each policy at its own terminal θ , and both policies on shared banks at $\theta \in \{38^\circ, 47^\circ, 50^\circ\}$. The shared-bank branches are what make the counterphased difference-in-differences interpretable, and $\theta = 50^\circ$ matches the difficulty at which the $d_{\text{mem}}=64$ parent was measured.

The counterphased estimand is

$$\text{DiD}(m) = \frac{1}{2} \left[(m_0 - m_3)|_{\text{high}=0} - (m_0 - m_3)|_{\text{high}=3} \right], \quad (1)$$

which cancels any location-intrinsic bias shared by the two policies. Positive $\text{DiD}(d')$ means each policy discriminates better at *its own* high-priority location.

4 Results

4.1 Learning and curriculum depth

Both runs learned and both stalled (Figure 1). The loc-0 lineage descended to $\theta = 47^\circ$ by iteration 5,839 and then advanced no further through iteration 19,516. The loc-3 lineage descended faster and further, reaching 41° by iteration 4,451 and 38° at 12,577. Neither approached the 8° floor, and both hovered near, but mostly below, the 0.85 advancement threshold — rolling correct rates settled around 0.80–0.82. The primary hypothesis of the design (that doubling d_{mem} “improves . . . curriculum depth”) is not supported by these partial runs: the deeper lineage here, 38° , is the one that ran *fewer* iterations.

4.2 Per-location signal detection

Table 2 gives the frozen-policy SDT decomposition at each policy’s own terminal θ . Counterphasing is *sign-correct in both lineages and for both metrics*: each policy has both higher d' and higher c at its own high-priority location. The magnitudes, however, are strikingly asymmetric — the loc-0 lineage shows $\Delta d' = +0.433$ while the loc-3 lineage shows only $\Delta d' = +0.053$ (CI includes zero). This is not a difficulty artifact: on the shared $\theta = 47^\circ$ bank the same gap persists (+0.460 versus +0.172), as it does at 50° (+0.348 versus +0.114). The positive group-level sensitivity effect is thus carried disproportionately by one of the two lineages, which is a real weakness of the counterphase in this pair.

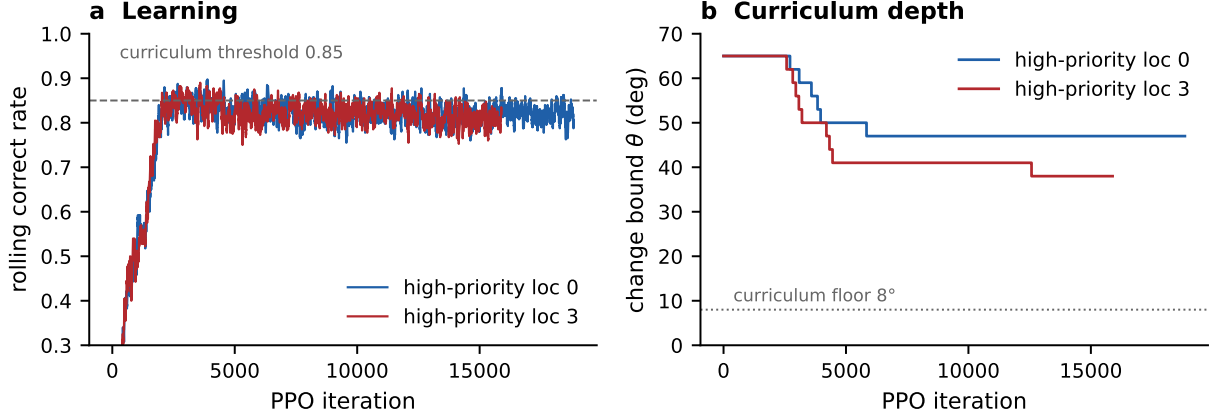


Figure 1: Training trajectories of the two rescued runs. (a) Rolling correct rate. (b) Curriculum change bound θ , a step function decremented by 3° whenever a 1000-trial window exceeds 0.85 correct. Both lineages stall well above the 8° floor.

Table 2: Frozen-policy SDT at each policy’s own terminal θ , trained mnemonic noise, sampled actions, 2,000 trials per change-status per location. * marks the high-priority (condition) location. The $d_{\text{mem}}=64$ parent is the terminal loc-3 checkpoint measured at $\theta = 50^\circ$.

Policy	loc	d'	c	HR	FA	n/status
$d_{\text{mem}}=128$ loc 0	0*	2.104	0.373	0.751	0.077	1999
(iter 18,799, $\theta=47^\circ$)	3	1.671	0.093	0.771	0.177	2000
$d_{\text{mem}}=128$ loc 3	0	2.007	0.409	0.724	0.079	2000
(iter 15,849, $\theta=38^\circ$)	3*	2.059	0.534	0.690	0.059	2000
$d_{\text{mem}}=64$ loc 3	0	1.625	0.062	0.774	0.191	2000
(iter 19,999, $\theta=50^\circ$)	3*	2.049	0.551	0.682	0.058	1999

Repeating the own- θ branch with mnemonic noise *disabled* changes almost nothing ($\Delta d' = +0.445$ and $+0.087$; $\Delta c = +0.295$ and $+0.126$), so neither the sensitivity effect nor the criterion shift is an artifact of noise injection at measurement time.

4.3 The counterphased test

The sensitivity result is robust: positive point estimate, CI excluding zero, at all three difficulties (Figure 2). The specificity result fails, though narrowly in absolute terms — the criterion CIs *straddle* the bound (lower limits 0.179–0.190 lie inside ± 0.2 ; upper limits 0.276–0.291 lie outside), so the equivalence test, which requires the entire interval inside the band, is not satisfied. The point estimates sit only 0.03–0.04 above the bound.

4.4 The criterion shift is reward-optimal, not a failure of learning

The sensitivity session assigns the high-priority location a mean reward of 5 with a hit:correct-rejection ratio of 0.7, and the low-priority location a mean reward of 1 with a ratio of 1.1. Expanding the ratios gives per-location payoffs of $R_{\text{hit}} = 4.118$, $R_{\text{CR}} = 5.882$ at the high location and $R_{\text{hit}} = 1.048$, $R_{\text{CR}} = 0.952$ at the low. Misses and false alarms pay zero and $P(\text{change}) = 0.5$ exactly, so

Table 3: Counterphased difference-in-differences (Equation 1) on shared trial banks, with 5,000-draw bootstrap CIs. The sensitivity target is met at every difficulty; the specificity target is missed at every difficulty. The final column is the reward-optimal criterion shift predicted from the task’s own reward table (Section 4.4), computed from the measured per-location d' .

	$\Delta d'_{\text{DiD}}$ (CI)	Δc_{DiD} (CI)	strict	Δc predicted
$\theta = 38^\circ$	+0.239 [+0.140, +0.335]	+0.227 [+0.179, +0.276]	fail	+0.237
$\theta = 47^\circ$	+0.316 [+0.216, +0.420]	+0.241 [+0.190, +0.291]	fail	+0.212
$\theta = 50^\circ$	+0.231 [+0.135, +0.330]	+0.228 [+0.179, +0.277]	fail	+0.213

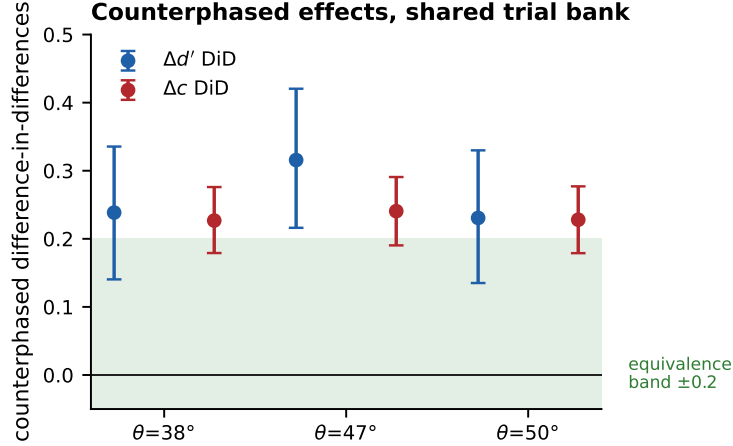


Figure 2: Counterphased difference-in-differences at three difficulties. The sensitivity effect (blue) clears zero everywhere. The criterion cross-effect (red) sits just above the ± 0.2 equivalence band (green) everywhere.

the reward-optimal likelihood-ratio criterion at a location is simply

$$\beta^* = \frac{P(\text{no change})}{P(\text{change})} \cdot \frac{R_{\text{CR}} - R_{\text{FA}}}{R_{\text{hit}} - R_{\text{miss}}} = \frac{R_{\text{CR}}}{R_{\text{hit}}} = \frac{1}{\text{H:CR ratio}}, \quad (2)$$

and for equal-variance Gaussian SDT the corresponding criterion is $c^* = \ln \beta^* / d'$. The high location therefore *should* be run more conservatively than the low one, by

$$\Delta \ln \beta^* = \ln(1/0.7) - \ln(1/1.1) = 0.452. \quad (3)$$

This is the crux. The reward table *mandates* a criterion difference; it is not a nuisance the agent could learn away. Converting Equation 3 into the criterion units of the equivalence bound, using each policy’s measured per-location d' , predicts $\Delta c_{\text{DiD}} = +0.213$ to $+0.237$ across the three difficulties. The observed values are $+0.227$, $+0.241$, $+0.228$ — agreement to within 0.03 (Table 3, Figure 3). These policies are sitting essentially exactly where the reward schedule tells them to sit.

Two consequences follow.

First, the specificity target is unreachable at the sensitivity these agents achieve. Setting $c^* \leq 0.2$ with a common d' at both locations requires

$$d' \geq \frac{0.452}{0.2} = 2.26, \quad (4)$$

whereas the measured values are 1.67–2.10 (Table 2). A perfectly reward-rational agent would fail this experiment’s specificity test. No increase in memory width, training length or curriculum

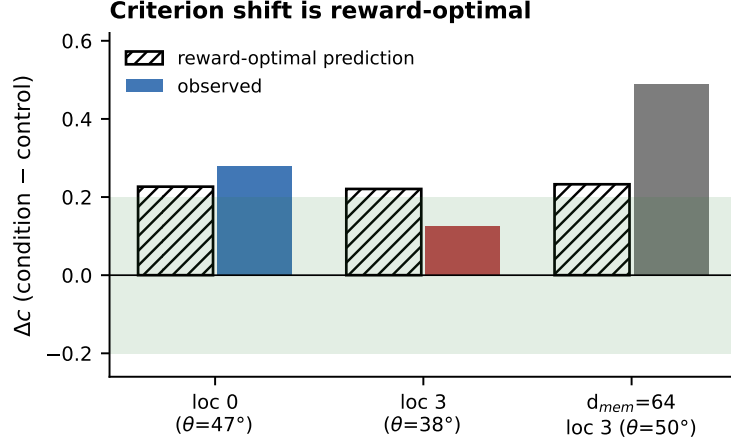


Figure 3: Observed criterion shift against the reward-optimal prediction $\Delta c^* = \ln \beta^* / d'$ computed from the measured per-location d' . The $d_{\text{mem}}=64$ parent overshoots the optimum by $2.1\times$; the $d_{\text{mem}}=128$ pair brackets it ($1.23\times$ and $0.57\times$). The optimum itself lies just outside the ± 0.2 band.

depth fixes that, because the requirement is a property of the payoff matrix, not of the policy — unless the agent were pushed to $d' > 2.26$, at which point the same reward table would become compatible with the bound.

Second, doubling the recurrent width did improve criterion calibration, which is a real and previously untested effect. The $d_{\text{mem}}=64$ parent produced $\Delta c = +0.489$ against a reward-optimal $+0.233$ — an overshoot of $2.1\times$, i.e. it was substantially *more* conservative at its high-value location than payoff maximisation warrants. The $d_{\text{mem}}=128$ policies give $1.23\times$ (loc 0) and $0.57\times$ (loc 3), bracketing the optimum, with the counterphased average landing on it. On the matched $\theta = 50^\circ$ bank the loc-3 cell moved from $\Delta c = +0.489$ at $d_{\text{mem}}=64$ to $\Delta c = +0.142$ at $d_{\text{mem}}=128$, i.e. from far outside the equivalence band to comfortably inside it — but its sensitivity effect shrank in step, from $\Delta d' = +0.425$ to $+0.114$.

5 Verdict

In one sentence: **the sensitivity half of the LM effect reproduced, the criterion specificity did not, and the reason it did not is that this reward schedule makes criterion invariance incompatible with reward-rational behaviour at the sensitivity these agents reach.**

The falsifier written into the experiment manifest — “the sensitivity gain remains coupled to a criterion shift outside the preregistered equivalence band” — is therefore met, and the d_{mem} capacity hypothesis is not rescued by these runs. But the falsifier fires for a reason that is external to the manipulation under test, so it should be read as a verdict on the *design*, not on memory capacity.

6 What to change before running this again

1. **Titrate the reward table.** Hold the hit:correct-rejection ratio *equal* across locations and vary only the mean reward, so that $\Delta \ln \beta^* = 0$ and the reward-optimal Δc is zero by construction. The sensitivity manipulation then acts through value/effort allocation without mandating a

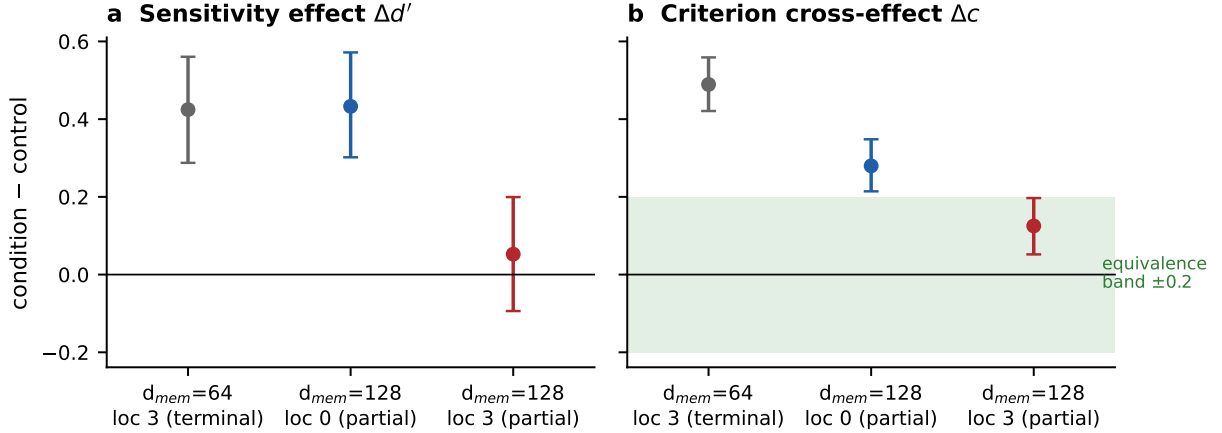


Figure 4: Per-policy condition-minus-control contrasts with bootstrap CIs. (a) The sensitivity effect is large in the loc-0 lineage and near zero in the loc-3 lineage. (b) The criterion cross-effect. The $d_{mem}=64$ parent sits far outside the equivalence band; both $d_{mem}=128$ policies are closer to it, and the loc-3 policy is inside it.

criterion difference, and $|\Delta c| \leq 0.2$ becomes a meaningful test rather than an arithmetic impossibility. This is the single change with the largest effect on whether the experiment can succeed.

2. **Report the reward-optimal criterion alongside the observed one** in the standard assay output. Both this run and its parent are far more interpretable as ratios to c^* than as raw Δc values; without it, the $d_{mem}=64$ result reads as a simple failure rather than as a $2.1\times$ overshoot.
3. **Fix the curriculum stall before adding capacity.** Both lineages parked well above the 8° floor at ≈ 0.80 – 0.82 correct against a 0.85 threshold. Sensitivity is the binding constraint on specificity here — $d' \geq 2.26$ would by itself bring the reward-optimal Δc inside the band — so effort spent reaching deeper θ is also effort spent making the specificity test passable.
4. **Run the criterion lineage.** Without it there is no double dissociation to claim, only one half of one.
5. **Investigate the lineage asymmetry.** A $8\times$ difference in $\Delta d'$ between two runs differing only in `--high-loc` is large enough to warrant seeds beyond seed 0 before the counterphased DiD is trusted as a group estimate.

7 Limitations

- **Partial checkpoints.** 94% and 79% of the training contract. The terminal-iteration validators were bypassed. The comparison against the $d_{mem}=64$ parent is therefore partial-versus-terminal, which flatters neither side in a predictable direction.
- **Single seed.** Seed 0 only, one policy per condition location. All CIs are trial-resampling intervals within a fixed pair of policies; they do not cover seed-to-seed variability, which the lineage asymmetry suggests is substantial.
- **Different terminal difficulties.** The two lineages stalled at 47° and 38° . This is handled by the shared-bank branches, and the DiD is stable across 38 – 50° , but no shared bank is

Table 4: Effect-by-effect verdict against the preregistered targets.

Target	Outcome	Evidence
Positive counterphased d' DiD	reproduced	+0.23 to +0.32, CI excludes zero at $\theta = 38, 47, 50^\circ$
Sign-correct counterphasing in both lineages	reproduced	Both policies have higher d' and higher c at their own high-priority location
Symmetric counterphase	partial	$\Delta d' = +0.433$ (loc 0) versus +0.053 (loc 3) at own θ ; gap persists at matched difficulty
Criterion cross-effect inside $ \Delta c \leq 0.2$	not reproduced	$\Delta c_{\text{DiD}} \approx +0.23$, CI upper limit 0.276–0.291
Strict behavioural dissociation	not achieved	Fails at every difficulty and for every individual policy
LM <i>double</i> dissociation	not testable here	Criterion lineage unsupported in this design; only sensitivity cells were run

simultaneously “terminal” for both policies.

- **Parent comparison is single-cell.** The $d_{\text{mem}}=64$ reference is one loc-3 checkpoint evaluated on its own bank at $\theta = 50^\circ$ with a different evaluation seed; there is no $d_{\text{mem}}=64$ loc-0 cell, so no parent DiD exists.
- **Equal-variance SDT.** $c^* = \ln \beta^* / d'$ assumes the equal-variance Gaussian model. Unequal variance would shift the predicted values somewhat, though not enough to move a 0.45 log-odds mandate inside a 0.2 band at $d' \approx 2$.
- **Not a neural replication.** This is a behavioural SDT analogue. Nothing here speaks to the V4 modulation results that motivate the original paper.

8 Artifacts

All inputs and outputs are under the rescue root

C:\Users\jomor\runpod_rescue\20260817. loc0/ and loc3/ hold the rescued run directories (checkpoint, full `metrics.csv`, supervisor log, manifests, launch and deployment contracts, pod provenance including `pip freeze`), `eval_tree/` holds the exact pod source plus the assay driver `run_partial_sdt.py` and its output `results/partial_sdt_results.json`, and `report/` holds `analyze_and_plot.py`, `analysis_summary.json`, the figures and this document. The assay is reproducible with

```
python run_partial_sdt.py --trials 2000 --bootstrap-draws 5000
--common-thetas 38 47 50
```

Checkpoints were written under NumPy 2 and require the `numpy._core` module alias when loaded under NumPy 1; the driver installs it.